

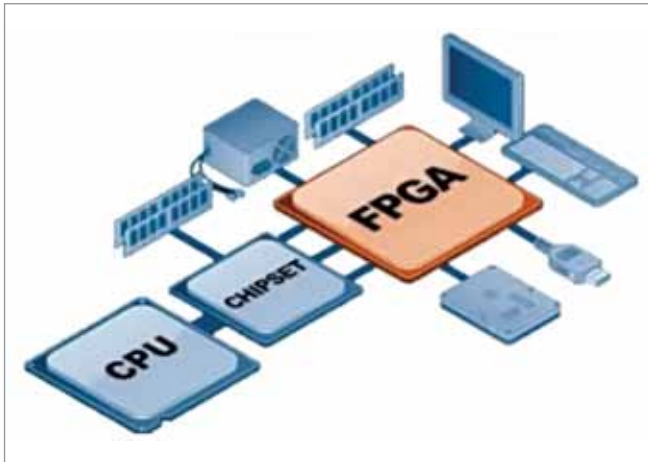


Fig. 4: Connection of an FPGA

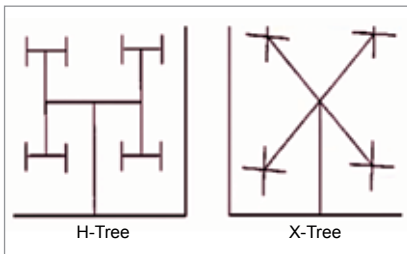


Fig. 5: Clock given at interaction

be routed properly so that it reaches every flip-flop at the same time to prevent clock skew. For that purpose, there are H-tree and X-tree types of circuit. In H-tree as well as X-tree, the clock is given at the intersection (of the letter H or X) as shown in Fig. 5. The flip-flops are placed on the edges, allowing the clock to reach simultaneously without any clock skew.

Next come power routing and ground (GND) routing. In ASIC, there are different layers for power and ground channels, which are connected using a via. The drilling of a via connection gives different chip layers for power routing and ground routing.

The next challenge is crosstalk. If the distance between two parallel signal lanes is too small, the signals may interfere with each other. Therefore, a specific distance must be maintained. Then we can download the bitstream into the FPGA and run it. The issue of crosstalk exists in ASIC but not in FPGA.

Since there are no flip-flop problems in an FPGA, its utilisation is quite high. But an ASIC should not heat up and should work in both low and high temperatures. If the flip-

flops are placed too close to each other, two different parts of the ASIC may have different temperatures, thus creating a design failure. So, we have to ensure that the circuit routing is done perfectly. We do not have to worry about this problem in FPGAs since the Vivado software

places them in small parts at different blocks of the FPGA.

Design challenges of FPGA

An ASIC is first tested in an FPGA before being launched. But one of the major drawbacks in this prototyping method is that the FPGA runs at a low speed as compared to an ASIC. ASIC runs at a fast clocking speed, which cannot be matched by an FPGA.

The next drawback is the power consumption. ASIC can operate on very low power as compared to an FPGA.

Also, latches cannot be accommodated in the FPGA since everything is present in the form of flip-flops and LUTs. But latches can be easily implemented in ASICs.

Other prototyping methods

Emulators can be used as a prototyping method and help in reducing the time to market even further than FPGAs. However, these are expensive and were launched just a few years ago in the market. FPGAs have been there for many years.

For thousands of signals and hundreds of lines of HDL code, emulators provide a full signal debug for our design, allowing us to precisely locate a bug in a particular part of the code.

One of the emulators is the Veloce emulator provided by Mentor Graphics for prototyping an ASIC. **EFY**

The article is based on the talk 'Rapid Prototyping Solutions: The Role Of FPGAs' by Abhinav Behl, RTL Designer at LogicFruit Technologies, which was presented at February edition of the Tech World Congress 2021.

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